PRACTICAL ASSESSMENT: AI-ASSISTED CUSTOMER SENTIMENT ANALYSIS

Platform: Microsoft Excel and approved Microsoft 365 Copilot **Suggested time:** 90–120 minutes **Total:** 100 marks

Dataset: Takaful_Customer_Sentiment_Student_Dataset.xlsx (Customer_Feedback worksheet; 800 synthetic records).

BUSINESS SCENARIO

You are a customer experience analyst for a fictional takaful provider. Management wants to understand customer satisfaction, recurring complaints and differences between service channels and branches. You have received 800 fictional customer feedback records in English and Bahasa Malaysia. Your task is to use Excel and Copilot critically to classify feedback, build a dashboard and recommend improvements.

INSTRUCTIONS

• Use only the supplied fictional dataset. Do not upload real customer information to any unapproved service.

• Retain the original data. Perform calculations in additional columns or worksheets.

• You may use Copilot to suggest classifications, formulas and summaries, but verify its output. Availability of row-by-row Copilot operations varies; if necessary, classify a representative sample of at least 100 records.

• Include screenshots or a short log of at least two Copilot prompts and describe how you checked the responses.

• Where percentages are reported, clearly state the denominator (number of feedback records).

PART A — DATA EXPLORATION AND QUALITY CHECK (15 MARKS)

A1 (5 marks): Describe the dataset, including the number of records, fields, branches, products, channels and languages. Identify any missing values or duplicates.

A2 (5 marks): Create a summary table of feedback count and average numerical rating by branch.

A3 (5 marks): Explain two reasons why a numerical rating may not accurately represent the sentiment expressed in a written comment.

PART B — AI-ASSISTED SENTIMENT CLASSIFICATION (25 MARKS)

B1 (5 marks): Define clear classification criteria for Positive, Negative, Neutral and Mixed sentiment. Explain how you will handle feedback that contains both praise and complaints.

B2 (15 marks): Use Copilot-assisted analysis to classify at least 100 feedback records, including examples in both English and Bahasa Malaysia. Populate Your_Sentiment in Excel and show the count and percentage of each sentiment class in the analysed sample.

B3 (5 marks): Present three difficult or ambiguous comments, their Feedback_IDs, the assigned sentiments and brief justifications. Identify any cases where you disagreed with Copilot.

PART C — COMPLAINT THEME EXTRACTION (20 MARKS)

C1 (10 marks): For the same analysed sample, assign one primary theme to each comment from: Processing Delay, Staff Service, Digital Platform, Documentation, Claims, Product Clarity, Payment or Communication. Populate Your_Primary_Theme.

C2 (5 marks): Among Negative and Mixed comments, identify the three most frequent primary themes. Provide counts and percentages using an explicitly stated denominator.

C3 (5 marks): Compare at least two themes across two branches or channels. Explain whether the observed differences could reflect small sample sizes or differences in feedback volume.

PART D — EXCEL DASHBOARD (20 MARKS)

D1 (15 marks): Create an Excel dashboard containing: (i) sentiment distribution chart, (ii) top complaint themes chart, (iii) a branch or channel comparison and (iv) at least two KPI measures. Label all charts and state the sample size.

D2 (5 marks): Write a concise interpretation of two important patterns visible in the dashboard. Avoid claiming that correlation establishes causation.

PART E — VALIDATION AND MANAGEMENT RECOMMENDATIONS (20 MARKS)

E1 (8 marks): Manually audit 40 classified comments, including Mixed comments and both languages. Record how many AI-assisted labels you agree with and calculate agreement rate = agreements / 40 × 100%. Discuss at least two disagreements or uncertain cases.

E2 (7 marks): Recommend two practical actions management should prioritise, supported by specific counts, percentages or examples from your analysis.

E3 (5 marks): Write a 120–180 word executive summary explaining key findings, recommended actions and two limitations of AI-based sentiment analysis.

REQUIRED SUBMISSIONS

1. Completed Excel workbook with classifications, calculations and dashboard.
2. A short report (PDF, maximum 3 pages) containing answers to Parts A–E, including charts and audit results.
3. At least two Copilot prompts and a brief explanation of how their outputs were verified.

MARKING SUMMARY

Part	Focus	Marks
A	Data exploration and quality	15
B	Sentiment classification	25
C	Complaint themes	20
D	Dashboard	20
E	Validation and recommendations	20
	TOTAL	100

Important: The reference labels are not provided to students. Your score depends on a defensible method, reproducible calculations and critical evaluation of Copilot, not merely copying an AI-generated answer.